Central Community Development Journal
https://www.acadlore.com/journals/CCDJ

Cohort-Stratified Evaluation of Participatory Training on Generative Fertilization and Fruiting-Booster Technologies Among Mangosteen Farmers

Paniran[1], Kurniawan Yuniarto[2], M. Azhar Mustafid[2], Reza Kusuma Nurrohman[2*], Sudirman[3]

[1] Department of Electrical Engineering, University of Mataram, 83125 Mataram, Indonesia
[2] Department of Agricultural Engineering, University of Mataram, 83125 Mataram, Indonesia
[3] Lingsar Agricultural Extension Office (BPP), Department of Agriculture of West Lombok, 83371 Lingsar, Indonesia

[*] Correspondence: Reza Kusuma Nurrohman (rezakusuman@unram.ac.id)

Received: 04-29-2026 **Revised:** 06-12-2026 **Accepted:** 06-20-2026

Citation: Paniran, Yuniarto, K., Mustafid, M. A., Nurrohman, R. K., & Sudirman. (2026). Cohort-stratified evaluation of participatory training on generative fertilization and fruiting-booster technologies among mangosteen farmers. *Cent. Community Dev. J.*, *6*(2), 99–117. https://doi.org/10.56578/ccdj060203.

Abstract: Mangosteen (*Garcinia mangostan*a L.) is a high-value crop in Indonesia, but reproductive-stage nutrient management remains challenging for smallholder farmers. This community-based study evaluated short-term changes in self-reported perceived knowledge and readiness-related responses following participatory training on generative fertilization and fruiting-booster technologies. Twenty-one mangosteen farmers received classroom instruction, interactive discussion, hands-on demonstrations, and supervised field training. Seven farmer-group leaders completed prospective pre- and post-training ratings, whereas 14 other participants provided post-training ratings and retrospectively reconstructed pre-training ratings after the intervention. The cohorts were analyzed separately using exact two-sided Wilcoxon signed-rank tests. Holm adjustment was applied to the six principal domain-by-cohort composite comparisons, and item-level analyses were exploratory. In the prospective cohort, generative-fertilization knowledge, fruiting-booster knowledge, and readiness-related composite scores increased by 2.07, 2.20, and 2.00 points, respectively; the corresponding increases in the retrospective cohort were 1.86, 1.69, and 1.54 points. All six comparisons remained significant after Holm adjustment (prospective, $p = 0.047$; retrospective, $p < 0.001$). Mean satisfaction scores were 4.77 ± 0.29 and 4.60 ± 0.38, respectively, both classified as very high. Participatory training was therefore associated with higher short-term self-reported ratings and high satisfaction. However, the uncontrolled small-sample design, retrospective ratings, and reliance on self-report preclude causal claims or conclusions about objective competence, sustained adoption, or farm outcomes.

Keywords: Mangosteen; Participatory agricultural extension; Farmer training; Perceived knowledge; Readiness-related responses; Generative fertilization; Fruiting-booster technology

1. Introduction

Mangosteen (*Garcinia mangostana* L.) is an economically important perennial tropical fruit with a distinctive domestication history (Goenaga & Rivera, 2005; Nazre, 2014; Yao et al., 2023). Indonesia is a major producer, and the crop supports rural livelihoods and local horticultural economies.

Production is constrained by a long juvenile period, irregular flowering, seasonal yield variation, and limited access to updated management guidance, particularly among smallholders (Osman & Milan, 2006). These constraints can reduce yield stability and farm profitability.

Reproductive-stage nutrient management influences flowering, fruit set, and fruit development. Phosphorus supports energy transfer and reproductive development, whereas potassium contributes to carbohydrate transport, enzyme activation, osmotic regulation, and fruit enlargement (Wang et al., 2013; Zörb et al., 2014). Because climate and rainfall affect nutrient availability and flowering phenology, local agroclimatic conditions should also inform nutrient-management decisions (Nugroho et al., 2020).

Calcium, magnesium, and boron also support reproductive development through cell-wall integrity, photosynthesis, carbohydrate metabolism, and pollen function (Shireen et al., 2018). Reproductive-stage nutrition may be combined with flowering-induction and biostimulant practices, although successful use depends on farmers' understanding and adaptation of these practices to local conditions (Calvo et al., 2014; du Jardin, 2015; Rouphael & Colla, 2020).

Smallholder adoption of agricultural innovations is often constrained by limited technical knowledge, extension access, experiential learning, and institutional support (Ruzzante et al., 2021). Knowledge alone may not change behavior without practical learning, social interaction, and continued technical support (Anderson & Feder, 2007). Participatory capacity building may therefore help connect technical recommendations with implementation.

Sustainable agricultural development also depends on farmers' capacity to make decisions, solve production problems, and collaborate within their communities (Christens, 2012). Community-based extension can strengthen local human capital, networks, and collaborative learning, although these broader outcomes require evidence beyond immediate training responses (Firmansyah et al., 2024; Pithakpol et al., 2025; Sukayat et al., 2023).

Participatory learning engages farmers in discussion, practice, reflection, and collaborative problem-solving rather than treating them as passive recipients of information (Pretty, 1995). By combining scientific guidance with local experience, it can support learning and engagement with agricultural innovations (Davis et al., 2012; Klerkx et al., 2012). Evidence of longer-term adoption, productivity, or community empowerment, however, requires direct follow-up.

A preliminary needs assessment in Batu Mekar Village combined orchard observations with informal discussions with farmers and extension officers. Farmers reported limited structured training in reproductive-stage nutrient management, and existing fertilization practices focused mainly on vegetative growth. These findings informed practical training on generative fertilization, flowering induction, and fruiting-booster technologies.

Although extension programs commonly evaluate knowledge and engagement with agricultural innovations (Anderson & Feder, 2007; Davis et al., 2012), evidence from participatory training in perennial horticultural systems such as mangosteen remains limited. Short-term evaluations should distinguish self-reported knowledge and readiness from actual competence, adoption, and community-level change.

Behavioral studies examine farmers' adoption intentions (Karbo et al., 2024; Nor Diana et al., 2024), but this program evaluated only three immediate outcomes: self-reported perceived knowledge, readiness-related responses, and participant satisfaction. It did not assess implementation, knowledge retention, empowerment, community capacity, or sustainable development; these outcomes require longitudinal and objective measurement.

Accordingly, this study evaluated participatory training on generative fertilization and fruiting-booster technologies for mangosteen farmers in West Lombok. The analysis examined short-term changes in self-reported perceived knowledge and readiness-related responses separately for prospective and retrospective assessment cohorts and described post-training satisfaction.

2. Methodology

2.1 Location and Participants

The program was conducted in Batu Mekar Village, West Lombok Regency, West Nusa Tenggara, Indonesia, where mangosteen (*Garcinia mangostana* L.) is an important smallholder crop (Figure 1). Twenty-one active mangosteen farmers managing productive orchards were purposively recruited through the village administration and local farmer group. All eligible farmers identified were invited; participation was voluntary, and no financial incentives were provided.

2.2 Program Design

The community service program was implemented through four sequential stages: (1) preliminary assessment, (2) training preparation, (3) capacity-building intervention, and (4) evaluation and reflection.

2.2.1 Preliminary assessment

A preliminary needs assessment combined direct observation of orchard management with informal discussions involving the Bina Mandiri Farmer Group and local agricultural extension officers.

The assessment covered reproductive-stage nutrient management, flowering induction, fruiting-booster use, routine fertilization, previous training, and production constraints.

Farmers reported practices focused mainly on vegetative growth and limited perceived knowledge of reproductive-stage management. These findings informed the training objectives, materials, and demonstrations.

2.2.2 Training preparation

Training materials were developed based on the identified farmer needs and focused on two major topics: (1)

generative fertilizer production and application and (2) fruiting booster production and application.

Educational materials included lecture presentations, practical guidelines, demonstration protocols, and evaluation questionnaires.

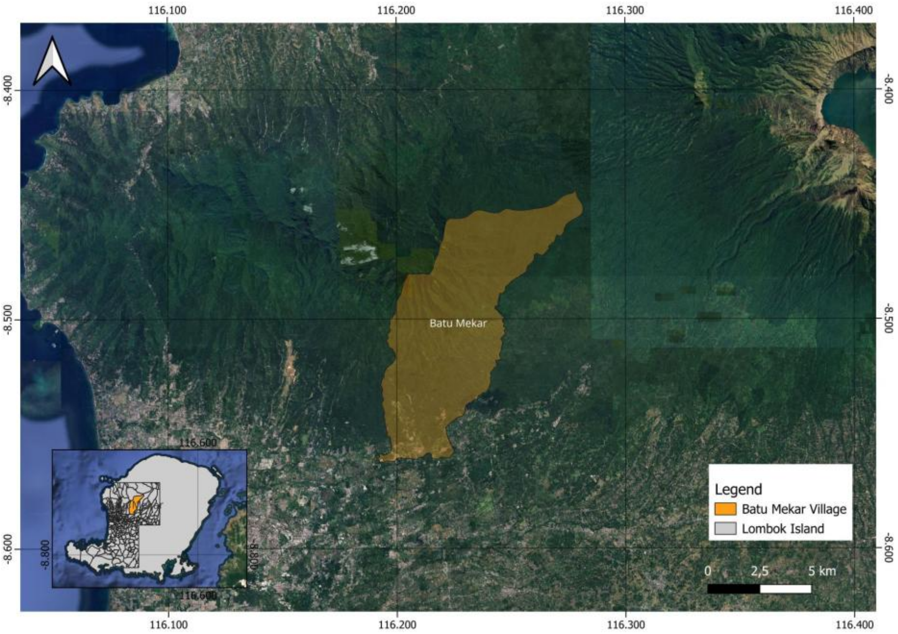

Figure 1. Location of the training site in Batu Mekar Village, West Lombok

2.2.3 Capacity-building intervention

The intervention ran from 20 May to 21 June 2026 and comprised a one-day workshop on theory and formulation preparation, followed by maturation and supervised field application in participants' orchards.

The evaluation included a prospective cohort of seven farmer-group leaders, who completed questionnaires before the workshop and after field application, and a retrospective cohort of 14 other participants, who provided post-training ratings and reconstructed pre-training ratings after the intervention.

On 20 May 2026, classroom instruction on reproductive-stage nutrient management was followed by discussion and hands-on preparation of the generative-fertilizer and fruiting-booster formulations. Participants prepared the formulations under instructor supervision, allowing direct practice and feedback (Davis et al., 2012; Kolb, 1984).

The generative fertilizer matured for 18 days, until 7 June, and the fruiting booster for approximately one month, until 20 June. Both were applied under supervision in participants' orchards on 21 June 2026.

The instructional team comprised Universitas Mataram lecturers, including a co-author specializing in fertilizer technology, and agricultural extension officers, combining academic and extension expertise (Anderson & Feder, 2007; Davis et al., 2012).

2.2.4 Evaluation and reflection

Structured questionnaires assessed self-reported perceived knowledge, readiness-related responses, and post-training satisfaction.

Pre-post analyses were stratified by prospective (n = 7) and retrospective (n = 14) assessment cohort; the latter provided both ratings after the intervention.

2.3 Training Materials and Workflow

Training covered the preparation and application of generative-fertilizer and fruiting-booster formulations through instruction, practical exercises, and supervised orchard application.

The workflow comprised classroom instruction, supervised formulation preparation, maturation (18 days and approximately one month, respectively), and orchard application. Participants performed the procedures while instructors provided guidance and feedback (Davis et al., 2012; Kolb, 1984).

Both formulations were presented as supplementary inputs, not replacements for recommended fertilization. Participants received standardized instructions on hygiene, fermentation, storage, dilution, timing, and application. The study did not test chemical composition, microbial identity or viability, fermentation quality, phytotoxicity, or agronomic efficacy.

2.3.1 Generative fertilizer module

The generative-fertilizer module demonstrated a fermented supplementary nutrient formulation containing mineral inputs, organic substrates, and a microbial inoculant.

Table 1 lists the quantities used per training batch and the intended functions of each ingredient; the product was not chemically characterized.

Table 1. Ingredients and intended functions of the training formulations

Ingredient	**Dose**	**Intended Role in Training Formulation**
	Generative fertilizer	
Nitrogen (N)	5 kg	Supports reproductive shoot growth and metabolic activity
Phosphorus (P)	4 kg	Promotes flowering initiation and root activity
Potassium (K)	5 kg	Improves flower retention, fruit set, and fruit quality
Calcium (Ca)	2 kg	Strengthens cell walls and supports fruit development
Magnesium (Mg)	2 kg	Essential component of chlorophyll and photosynthesis
Boron (B)	2 kg	Facilitates pollen germination and fruit set
Coconut water	30 L	Natural source of phytohormones and carbohydrates
Rice-washing water	30 L	Carbon source supporting microbial fermentation
Microbial inoculant	2 L	Accelerates organic matter decomposition and nutrient transformation
Organic fermentation additive	2 kg	Enhances microbial activity during fermentation
	Fruiting booster	
Ripe sapodilla fruit	6 kg	Natural carbohydrate source for fermentation
Milk	1.2 L	Provides amino acids and nutrients for microbial growth
Duck eggs	5 eggs	Source of proteins and amino acids
Honey	100 mL	Easily available sugars supporting fermentation
Immature dates	750 g	Natural sugars and micronutrients
Coconut water	5 L	Source of natural growth-promoting compounds
Rice-washing water	5 L	Supports beneficial microbial activity
Banana blossom	1 kg	Natural source of potassium and plant metabolites
Seed extracts (Mangosteen, avocado, and durian seeds)	1–2 seeds each	Additional plant bioactive compounds
Microbial inoculant	1 L	Initiates fermentation
Fermentation enhancer	100 mL	Stabilizes microbial fermentation

Participants sanitized containers with 100 mL of 70% alcohol, mixed the ingredients, added 2 L of microbial inoculant and 2 kg of Guren, and fermented the preparation in sealed containers for at least 18 days.

Participants were instructed to dilute 100 mL of the preparation in 15–16 L of water and apply it at least twice monthly from approximately four months before anticipated flowering, either as a foliar or soil spray or as a root-zone drench, while avoiding strong sunlight.

Alternative preparation procedures, dilution rates, and application schedules were not evaluated. All participants followed the same demonstrated formulation and application protocol. Containers were cleaned, sealed during fermentation, and stored in shaded or dark conditions.

No laboratory assessment of fermentation quality, microbial viability, chemical composition, or phytotoxicity was performed.

2.3.2 Fruiting booster module

The fruiting-booster module demonstrated a fermented complementary biostimulant formulation based on locally available organic substrates and a microbial inoculant.

Table 1 lists the ingredients, quantities used per training batch, and intended functions. The formulation was neither chemically characterized nor independently validated.

Participants cleaned and blended the ingredients, added the inoculant, left approximately 10% container headspace, sealed the mixture, and fermented it in shaded or dark conditions for at least one month.

Participants were instructed to dilute 10 mL of the formulation in 15–16 L of water and apply it evenly as a foliar spray at least twice between flowering and fruit ripening. All participants followed the same demonstrated formulation and application protocol.

No laboratory testing of microbial identity or viability, chemical composition, fermentation quality, phytotoxicity, or agronomic efficacy was conducted; alternative formulations and application regimes were not evaluated.

2.3.3 Participatory learning activities

Training combined lectures, discussion, question-and-answer sessions, practical demonstrations, and field application. Participants prepared and applied the formulations, discussed local production constraints, and related the procedures to their orchards.

2.3.4 Training workflow

Figure 2 summarizes the four program phases of prospective and retrospective assessment: preliminary assessment, preparation, training and field application, and short-term evaluation.

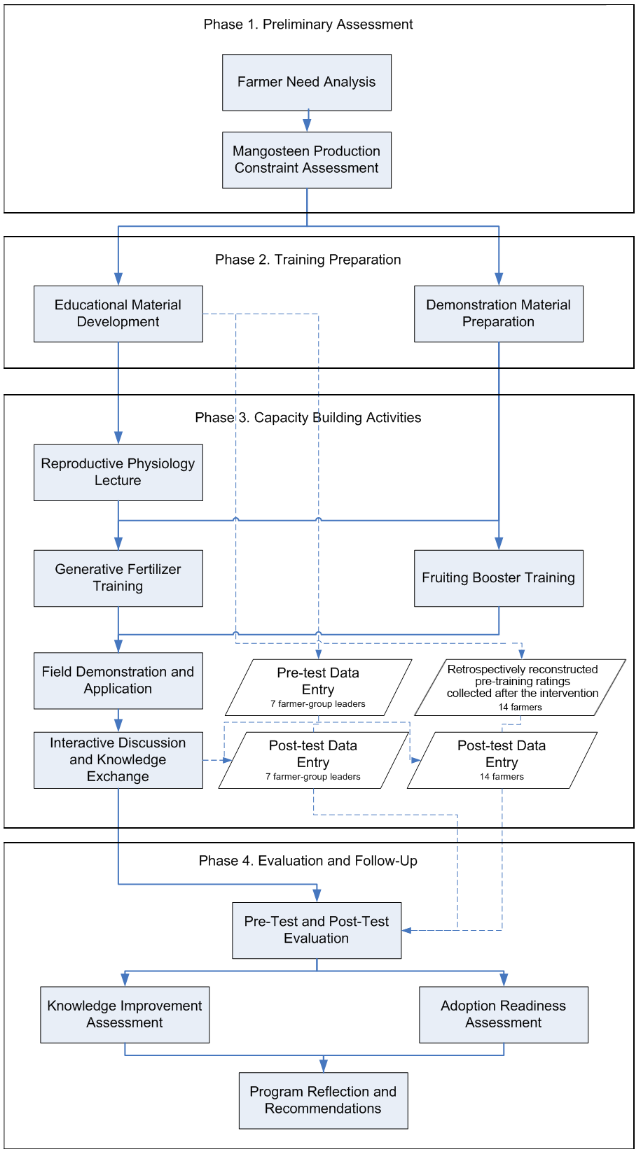

Figure 2. Community training workflow

Note: For the retrospective cohort, "pre-training" denotes ratings reconstructed after completion of the intervention and not a prospective baseline assessment.

2.4 Data Collection

Questionnaires were completed prospectively by seven farmer-group leaders immediately before the workshop on 20 May 2026 and after supervised field application on 21 June 2026. On 9 July 2026, the remaining 14 participants completed post-training ratings and retrospectively reconstructed pre-training ratings. The cohorts were analyzed separately.

The authors developed a 28-item paper questionnaire for this program from its learning objectives and relevant extension and adoption literature (Anderson & Feder, 2007; Davis et al., 2012; Meijer et al., 2015; Ruzzante et al., 2021). The administered GF4 and AR4 items inadvertently referred to *mangga* (mango). Although the authors report that *manggis* (mangosteen) was clarified during administration, both items were excluded. GF5 and AR5 were renumbered GF4 and AR4, leaving 26 retained items: eight participant-characteristic items, nine perceived-knowledge items, four readiness-related items, and five satisfaction items. Appendix B presents the retained, blank, de-identified analytical questionnaire, not the administered form.

One expert in horticultural science, agricultural extension, and community-based farmer training reviewed the draft questionnaire for relevance, clarity, local suitability, readability, response format, and ambiguity. The authors revised the wording accordingly. No quantitative content-validity assessment was conducted.

The instrument was developed for program evaluation rather than as a validated psychometric scale; internal consistency was therefore not assessed. All outcome items used a five-point Likert scale (1 = strongly disagree; 5 = strongly agree). Participant-level composites were the arithmetic means of retained items within each domain and were interpreted as descriptive indices.

The six cohort-stratified domain-composite comparisons formed the principal inferential family; item-level comparisons were exploratory. This post hoc reporting hierarchy and the statistical procedures are detailed in Section 2.5.

2.4.1 Participant characteristics

Participant characteristics included gender, age, education, farming experience, mangosteen area, productive tree count, annual fruit production, and prior agricultural training (Table 2).

Table 2. Demographic and farming characteristics of participants

Characteristic	Prospective	Retrospective
Total participants, *n*	7	14
Gender, *n* (%)		
Male	5 (71.4%)	2 (14.3%)
Female	2 (28.6%)	12 (85.7%)
Age (years), mean ± SD	37.9 ± 8.6	38.4 ± 9.0
Education level, *n* (%)		
Elementary school	1 (14.3%)	0 (0.0%)
Junior high school	2 (28.6%)	8 (57.1%)
Senior high school	2 (28.6%)	4 (28.6%)
Bachelor's degree	2 (28.6%)	2 (14.3%)
Farming experience (years), mean ± SD	15.3 ± 10.2	13.9 ± 6.6
Plantation area (ha), mean ± SD	0.28 ± 0.21	0.31 ± 0.18
Number of productive trees, mean ± SD	59.1 ± 24.7	62.4 ± 16.9
Average annual production (kg), mean ± SD	1428.6 ± 925.0	1450.0 ± 834.6
Previous Similar Training Experience, *n* (%)		
Yes	7 (100.0%)	3 (21.4%)
No	0 (0.0%)	11 (78.6%)

Note: SD = standard deviation. *n* = sample size.

2.4.2 Self-reported perceived knowledge and readiness-related responses

The questionnaire retained 13 outcome statements across three domains (Table 3): generative-fertilization perceived knowledge (four items), fruiting-booster perceived knowledge (five), and readiness-related responses (four).

Table 3. Retained questionnaire items by outcome domain

Domain	Code	Measurement Item
Perceived Knowledge of Generative Fertilization	GF1	I understand the difference between vegetative and generative fertilization.
	GF2	I understand the role of phosphorus (P) and potassium (K) in flowering and fruit formation.
	GF3	I understand the appropriate timing for generative fertilizer application.
	GF4	I understand the expected benefits of generative fertilization for fruit production.
Perceived Knowledge of Fruiting Booster Technology	FB1	I understand the function of fruiting boosters in stimulating flowering.
	FB2	I know the correct application method for fruiting boosters.
	FB3	I understand the risks associated with excessive fruiting booster application.
	FB4	I know the environmental conditions that influence flowering success.
	FB5	I understand how fruiting boosters can affect mangosteen yield and fruit quality.
Perceived Readiness to Adopt the Technology	AR1	I am confident that I can independently apply the techniques introduced during the training.
	AR2	I am willing to implement the introduced technology in my mangosteen orchard.
	AR3	I believe that the technology is economically feasible.
	AR4	I intend to recommend the technology to other farmers.

Note: Retained outcome items were rated with reference to the pre- and post-training periods on a five-point Likert scale (1 = strongly disagree; 5 = strongly agree). In the retrospective cohort, both ratings were collected after the intervention.

Generative-fertilization items covered the vegetative–generative distinction, phosphorus and potassium, application timing, and expected production benefits.

Fruiting-booster items covered function, application, excessive-use risks, environmental influences on flowering, and potential effects on yield and fruit quality.

Readiness-related items addressed confidence, willingness to implement, perceived economic feasibility, and intention to recommend. They were treated as distinct but related indicators, not a validated unidimensional scale.

Participant-level composites were arithmetic means of retained items within each domain; individual items were summarized as mean ± standard deviation (SD) at each time point.

Principal and exploratory inferential procedures are described in Section 2.5. The five-point Likert format was selected as a practical tool for community-based program evaluation (Joshi et al., 2015).

2.4.3 Participant satisfaction

Post-training satisfaction items assessed instructional clarity, practical usefulness, relevance, and overall experience. The five items covered the training materials, instructor explanations, hands-on demonstrations, relevance to farming, and overall satisfaction.

Responses used the same five-point scale and were summarized descriptively by item and as a participant-level five-item composite. Scores were classified as very high (4.21–5.00), high (3.41–4.20), moderate (2.61–3.40), low (1.81–2.60), or very low (1.00–1.80).

2.5 Data Analysis

Participant characteristics were summarized using frequencies, percentages, means, and SDs. Five-point item responses were summarized as mean ± SD, and participant-level domain composites were calculated as the arithmetic means of four generative-fertilization items, five fruiting-booster items, and four readiness-related items.

Paired analyses were stratified by prospective and retrospective cohort to avoid pooling ratings obtained through different pre-training assessment procedures.

Each of the six domain-by-cohort composite comparisons was evaluated using an exact two-sided Wilcoxon signed-rank test (Wilcoxon, 1945). Zero differences were excluded, tied absolute differences received midranks, and exact *p*-values were obtained by exhaustive enumeration of all possible sign assignments to the nonzero paired differences.

For each comparison, the positive-rank sum (W^+), exact two-sided *p*-value, mean change, and matched-pairs rank-biserial correlation (r_{rb}; range −1 to 1) were reported. Holm's procedure controlled the family-wise error rate across the six composite comparisons. Both unadjusted and adjusted *p*-values are shown. This reporting hierarchy was adopted post hoc during revision and was not prespecified.

The 13 retained items were also compared within each cohort using exact two-sided Wilcoxon signed-rank tests, yielding 26 secondary exploratory tests. These *p*-values were not adjusted for multiplicity and were not treated as confirmatory evidence.

Exact two-sided sign tests assessed the direction of change in each of the six composites; ties were recorded as zero and excluded from the sign counts. These tests were sensitivity analyses, not independent evidence of effectiveness.

Satisfaction was summarized descriptively by item and five-item participant-level composite and classified using the thresholds in Section 2.4.3. It was not included in pre–post inference.

Analyses were conducted in Python 3.13 using pandas, NumPy, and Python-based statistical routines; figures were generated with Matplotlib.

3. Results

3.1 Participant Profile

Twenty-one farmers were analyzed separately as a prospective cohort (n = 7) and a retrospective cohort (n = 14). According to Table 2, cohorts differed notably in gender composition: 71.4% of the prospective cohort were men, whereas 85.7% of the retrospective cohort were women. Mean ages were similar (37.9 ± 8.6 vs. 38.4 ± 9.0 years).

The prospective cohort had a broader distribution of educational attainment, whereas junior high school predominated in the retrospective cohort (57.1%). Farming experience, orchard area, productive tree count, and annual production were descriptively similar between cohorts; both comprised experienced smallholders.

All prospective participants had attended similar training previously, compared with 21.4% of retrospective participants. This imbalance may have influenced baseline perceptions but was not analyzed causally. Given these cohort differences and the distinct assessment procedures, the cohorts were not pooled.

3.2 Implementation of Training Activities

Training combined classroom instruction, discussion, hands-on formulation preparation, maturation, and supervised orchard application.

Lectures covered reproductive-stage nutrition, flowering and fruit development, generative fertilization, and the intended functions of the fruiting-booster formulation; Figure 3 shows the classroom and practical activities.

Participants prepared ingredients, managed fermentation, practiced application, and discussed local constraints such as inconsistent flowering, low fruit set, and variable productivity. These activities provided the context for the subsequent self-reported evaluations.

(a) (b)

Figure 3. Classroom and practical training activities: (a) nutrient-management lecture and (b) preparation of the training formulations

3.3 Changes in Self-Reported Perceived Knowledge

Participant-level composites for generative-fertilization and fruiting-booster perceived knowledge were analyzed separately by cohort (Table 4). Secondary item-level results are shown in Figure 4 and Figure 5, driven from data in Appendix Table A1 and A2.

Table 2. Principal domain-level comparisons by assessment cohort

Outcome	Cohort	n	Pre-Training	Post-Training	Mean	W^+	Exact p	Holm-Adjusted p	r_{rb}
Generative Fertilization	Prospective	7	2.93 ± 0.43	5.00 ± 0.00	+2.07	28	0.016	0.047	1.00
Generative Fertilization	Retrospective	14	2.93 ± 0.41	4.79 ± 0.35	+1.86	105	<0.001	<0.001	1.00
Fruiting Booster	Prospective	7	2.80 ± 0.26	5.00 ± 0.00	+2.20	28	0.016	0.047	1.00
Fruiting Booster	Retrospective	14	3.00 ± 0.48	4.69 ± 0.41	+1.69	105	<0.001	<0.001	1.00
Readiness-related responses	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	0.047	1.00
Readiness-related responses	Retrospective	14	3.14 ± 0.36	4.68 ± 0.41	+1.54	105	<0.001	<0.001	1.00

Note: Domain scores are participant-level means of the retained items. n = sample size; SD = standard deviation; W^+ denotes the sum of positive Wilcoxon signed ranks; r_{rb} denotes the matched-pairs rank-biserial correlation. Unadjusted exact p-values are two-sided; Holm adjustment was applied across the six principal domain-by-cohort comparisons. Zero differences were excluded from the Wilcoxon ranking.

In the prospective cohort, the generative-fertilization composite increased from 2.93 ± 0.43 to 5.00 ± 0.00 (mean change, +2.07; $W^+ = 28$; unadjusted exact $p = 0.016$; Holm-adjusted $p = 0.047$; $r_{rb} = 1.00$). In the retrospective cohort, the corresponding composite increased from 2.93 ± 0.41 to 4.79 ± 0.35 (+1.86; $W^+ = 105$; unadjusted exact $p < 0.001$; Holm-adjusted $p < 0.001$; $r_{rb} = 1.00$). The prospective fruiting-booster composite increased from 2.80 ± 0.26 to 5.00 ± 0.00 (+2.20; $W^+ = 28$; unadjusted exact $p = 0.016$; Holm-adjusted $p = 0.047$; $r_{rb} = 1.00$). The retrospective fruiting-booster composite increased from 3.00 ± 0.48 to 4.69 ± 0.41 (+1.69; $W^+ =$

105; unadjusted exact $p < 0.001$; Holm-adjusted $p < 0.001$; $r_{rb} = 1.00$).

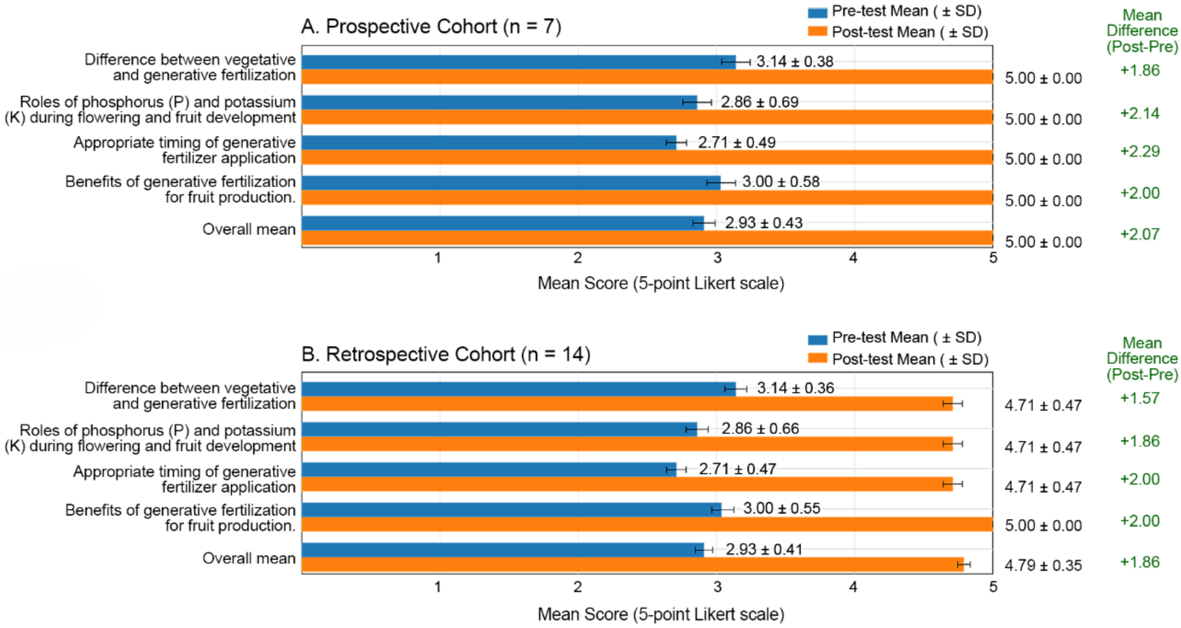

Figure 4. Exploratory item-level changes in generative-fertilization perceived knowledge

Exploratory item-level changes in generative-fertilization knowledge were positive in both cohorts. The largest increase was for application timing (GF3: prospective, +2.29; retrospective, +2.00), with GF4 also increasing by +2.00 retrospectively (Appendix Table A1; Figure 4).

For fruiting-booster knowledge, the largest increase in both cohorts concerned environmental conditions affecting flowering (FB4: prospective, +2.43; retrospective, +1.86; Appendix Table A2; Figure 5). Other retrospective item changes ranged from +1.43 to +1.71.

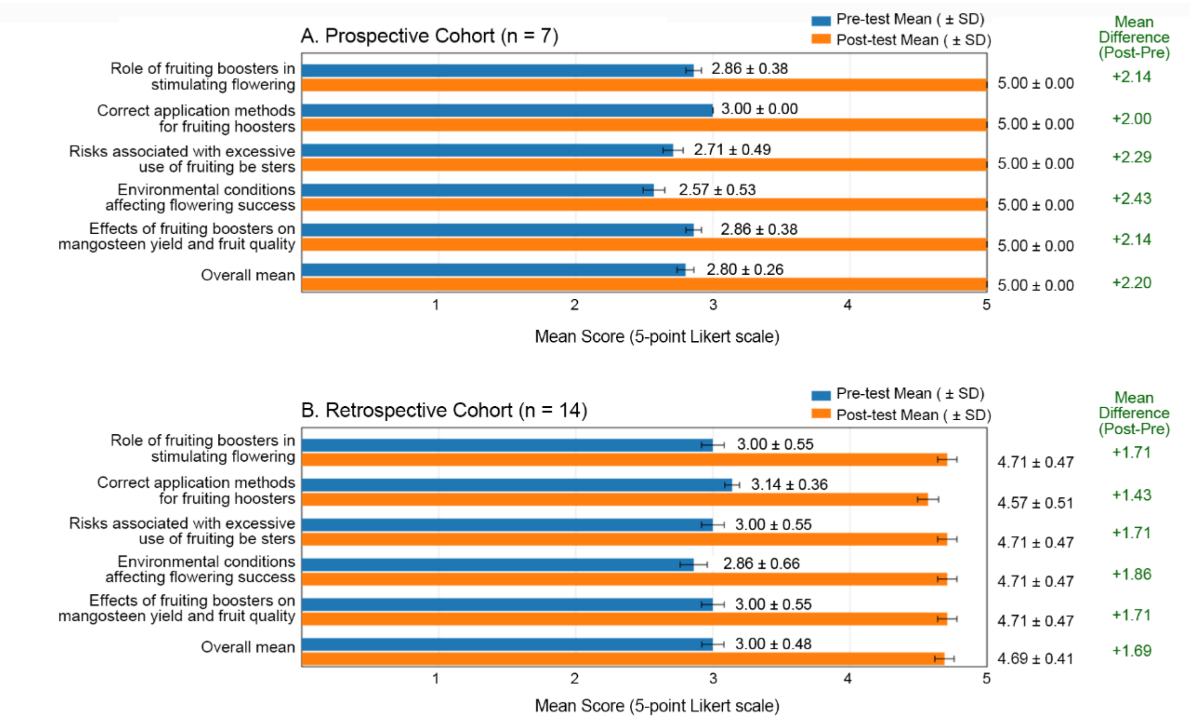

Figure 5. Exploratory item-level changes in fruiting-booster perceived knowledge

Sign-test sensitivity analyses showed positive changes for every participant in both knowledge domains (prospective, $p = 0.016$; retrospective, $p < 0.001$; Appendix Table A4).

Thus, both knowledge composites increased in both cohorts after Holm adjustment (Table 4). Because outcomes were self-reported and assessed shortly after training, they do not demonstrate objective competence, retention, or implementation.

3.4 Changes in Readiness-Related Responses

Readiness-related responses were summarized in a four-item participant-level composite and analyzed separately by cohort (Table 4); secondary item-level results appear in Figure 6 and Appendix Table A3.

Figure 6. Exploratory item-level changes in readiness-related responses

In the prospective cohort, the composite increased from 3.00 ± 0.00 to 5.00 ± 0.00 (+2.00; $W^{+} = 28$; unadjusted exact $p = 0.016$; Holm-adjusted $p = 0.047$; $r_{rb} = 1.00$; Table 4). In the retrospective cohort, it increased from 3.14 ± 0.36 to 4.68 ± 0.41 (+1.54; $W^{+} = 105$; unadjusted exact $p < 0.001$; Holm-adjusted $p < 0.001$; $r_{rb} = 1.00$; Table 4).

Exploratory item-level changes were +2.00 for all four prospective items. Retrospective changes ranged from +1.43 to +1.71 and were largest for willingness to implement the technology (AR2; Appendix Table A3; Figure 6).

Sign-test sensitivity analyses showed positive changes for every participant in both cohorts (prospective, $p = 0.016$; retrospective, $p < 0.001$; Appendix Table A4).

The prospective cohort reached the scale maximum on all four post-training items, whereas retrospective post-training means ranged from 4.57 to 4.86 (Appendix Table A3; Figure 6).

The composite is a program-evaluation index of related but distinct self-reported responses, not a validated adoption-readiness scale or evidence of subsequent adoption.

Both cohort comparisons remained significant after Holm adjustment, indicating higher short-term readiness-related ratings but not sustained implementation or behavioral change (Table 4).

3.5 Participant Satisfaction

Post-training satisfaction was analyzed descriptively by cohort (Table 5).

The five-item satisfaction composite was classified as very high in both cohorts: 4.77 ± 0.29 prospectively and 4.60 ± 0.38 retrospectively.

In the prospective cohort, training relevance and overall satisfaction each scored 5.00 ± 0.00; the mean values of other items ranged from 4.57 to 4.71 (Table 5).

In the retrospective cohort, overall satisfaction scored 5.00 ± 0.00, and other means ranged from 4.43 to 4.57.

Item means exceeded 4.40 in both cohorts (Table 5).

These ratings describe favorable immediate perceptions and do not establish instructional effectiveness, sustained engagement, or adoption.

Table 3. Participant satisfaction by assessment cohort

Code	Satisfaction Indicator	Prospective (n = 7)	Satisfaction Level	Retrospective (n = 14)	Satisfaction Level
TE1	The training material was easy to understand.	4.71 ± 0.49	Very high	4.57 ± 0.51	Very high
TE2	The instructors explained the material clearly.	4.57 ± 0.53	Very high	4.43 ± 0.51	Very high
TE3	The hands-on demonstrations helped me understand the material better.	4.57 ± 0.53	Very high	4.43 ± 0.51	Very high
TE4	This training was relevant to my farming activities.	5.00 ± 0.00	Very high	4.57 ± 0.51	Very high
TE5	Overall, I am satisfied with this training program.	5.00 ± 0.00	Very high	5.00 ± 0.00	Very high
	Overall satisfaction composite	4.77 ± 0.29	Very high	4.60 ± 0.38	Very high

Note: Values are mean ± standard deviation on a five-point Likert scale. Satisfaction categories were very high (4.21–5.00), high (3.41–4.20), moderate (2.61–3.40), low (1.81–2.60), and very low (1.00–1.80). n = sample size.

4. Discussion

4.1 Interpretation of Readiness-Related Responses

Both cohorts reported higher readiness-related composite scores after training, and both comparisons remained significant after Holm adjustment (Table 4). These results indicate short-term changes in perceived confidence, willingness to implement, economic feasibility, and intention to recommend, rather than verified adoption.

Exploratory item results were positive throughout, with retrospective changes largest for willingness to implement and confidence (Appendix Table A3). Because these tests were unadjusted, they provide descriptive detail only.

Participatory instruction, practice, and discussion may strengthen perceived readiness by connecting technical guidance with farmers' experience. However, the study did not measure decision-making autonomy, competence, implementation, or longer-term behavior; it therefore provides no direct evidence that empowerment occurred. The pattern is consistent with research linking participatory extension to farmer learning and engagement (Firmansyah et al., 2024; Pithakpol et al., 2025; Sukayat et al., 2023), but longitudinal evidence is needed to determine whether reported readiness persists or leads to use.

Sign-test sensitivity analyses confirmed that the direction of change was positive for every participant, but they do not provide independent evidence of adoption or empowerment (Appendix Table A4). The larger descriptive increase in the prospective cohort should not be interpreted as greater effectiveness because assessment mode was confounded with cohort characteristics, including participant role and prior training.

Overall, the readiness findings suggest a potential contribution to farmer capacity building, subject to objective and longitudinal verification.

4.2 Perceived Knowledge and Community Capacity

The program provided shared technical instruction on generative fertilization and fruiting-booster practices, and participants in both cohorts reported higher perceived knowledge after training. All four knowledge domain-by-cohort composites remained significant after Holm adjustment, with positive changes throughout (Table 4). These individual learning outcomes may support—but do not demonstrate—community capacity.

Exploratory item results identify content areas with the largest perceived changes but should not be interpreted as separate primary outcomes (Appendix Tables A1 and A2). Sign-test results confirmed a uniformly positive direction of participant-level change (Appendix Table A4). Training through an existing farmer group may provide a shared basis for discussion and interaction with extension staff. Nevertheless, the study measured individual self-reports, not collective knowledge, institutional capacity, group functioning, coordinated decision-making, or knowledge-sharing behaviour.

Participatory extension can contextualize technical knowledge within local experience and potentially support community learning (Labonté & Laverack, 2001; Laverack, 2006). Whether the observed individual changes contribute to collective learning, group capacity, or sustained community development requires follow-up.

4.3 Potential for Social Learning and Knowledge Diffusion

Discussion and practical work created opportunities for participants to exchange experiences with peers, trainers, and local stakeholders, consistent with social-learning processes based on observation, discussion, experimentation, and reflection (Ensor & Harvey, 2015; Reed et al., 2010).

Higher perceived-knowledge ratings in both domains show an immediate self-reported learning pattern, not evidence that knowledge was retained or shared beyond participants (Table 4). Exploratory item results indicate which content areas changed most but add no evidence of collective learning or diffusion (Appendix Tables A1and A2).

Intention to recommend increased in both cohorts, suggesting a possible pathway for interpersonal diffusion (Rogers, 2003). However, intention does not show that participants subsequently shared the information. The sensitivity analyses likewise confirmed a positive direction of participant-level change but did not assess knowledge retention or diffusion (Appendix Table A4).

The study did not track communication to other farmers, uptake by nonparticipants, or spread beyond the group. Knowledge diffusion therefore remains a hypothesis for future study. These findings align with participatory-extension studies reporting short-term gains in knowledge and engagement (Davis et al., 2012; Meijer et al., 2015), but longitudinal follow-up is required to assess competence, communication, adoption, and diffusion.

4.4 Implications and Limits

The observed changes and high satisfaction indicate a favorable short-term learning experience. Such human-capital development may support rural livelihoods by helping farmers evaluate management options (Scoones, 1998). However, because yield, fruit quality, production costs, profitability, and household outcomes were not measured, implications for rural livelihoods remain conceptual.

All six principal composite comparisons showed higher post-training ratings after Holm adjustment (Table 4). Satisfaction was very high in both cohorts (Table 5), indicating favorable immediate perceptions of training content and delivery. Adoption may depend on inputs, finance, labor, markets, risk, institutional support, and continued extension access (Knowler & Bradshaw, 2007; Ruzzante et al., 2021). Objective and longitudinal monitoring is needed to determine whether reported changes lead to implementation and farm outcomes.

Future evaluations should assess knowledge retention, implementation, orchard management, productivity, costs, profitability, and farmer-to-farmer diffusion across multiple seasons.

4.5 Study Limitations and Future Research

The findings should be interpreted in light of several limitations. First, the uncontrolled one-group design cannot attribute observed changes solely to training; testing, memory, response, maturation, or external effects may also have contributed (Shadish et al., 2002). The findings indicate association, not causal effectiveness.

Second, the purposive sample of 21 farmers from one village limits precision and generalizability, especially for the seven-person prospective cohort.

Third, immediate post-training assessment did not evaluate retention, sustained behavior, or implementation. Follow-up should use field observations, farm records, or other objective measures across multiple seasons.

Fourth, self-reported knowledge, readiness, and satisfaction are susceptible to social-desirability and response biases (Podsakoff et al., 2003) and do not establish competence or instructional effectiveness.

Fifth, seven farmer-group leaders completed prospective assessments, whereas 14 other participants reconstructed pre-training ratings after the intervention. Assessment mode was therefore confounded with participant role and other cohort characteristics, precluding attribution of between-cohort differences to assessment mode. Retrospective ratings are also vulnerable to recall, response-shift, consistency, and demand-characteristic biases.

Although all six composites changed positively, smaller retrospective-cohort *p*-values partly reflect the larger sample and do not indicate stronger design evidence (Table 4).

Sixth, satisfaction was measured only after training and therefore indicates perception, not change or prediction of later adoption.

Finally, the study did not verify implementation or agronomic outcomes; positive sign-test results establish only the direction of short-term self-reported change (Appendix Table A4).

Future studies should use larger and more diverse samples, comparison groups where feasible, longitudinal follow-up, and objective measures of implementation, agronomic and economic outcomes, adoption barriers, cost-effectiveness, and knowledge diffusion.

These priorities would clarify whether short-term self-reported changes persist and translate into adoption or farm-level benefits.

5. Conclusions

This study evaluated short-term self-reported outcomes of participatory training on generative fertilization and fruiting-booster technologies among 21 purposively selected mangosteen farmers in one West Lombok village.

Perceived-knowledge composites increased in both cohorts for both technology domains, and all four comparisons remained significant after Holm adjustment. These ratings do not measure objective knowledge or competence.

Readiness-related composites also increased in both cohorts after Holm adjustment, reflecting higher reported confidence, willingness, economic feasibility, and intention to recommend—not verified adoption or sustained behavior. Post-training satisfaction was classified as very high in both cohorts.

Overall, the program was associated with positive short-term self-reported changes and favorable perceptions, supporting its potential as an educational approach without establishing causal effects or farm-level benefits.

Interpretation is limited by the uncontrolled design, small purposive sample, single setting, cohort differences, retrospective ratings, and self-report. Longitudinal controlled studies with objective measures are needed to determine whether these changes persist and translate into competence, adoption, and agronomic or economic outcomes.

Author Contributions

Conceptualization, R.K.N., P., K.Y., and M.A.M.; methodology, R.K.N., K.Y., M.A.M., and S.; validation, R.K.N., K.Y., M.A.M., and S.; formal analysis, R.K.N.; investigation, R.K.N. and M.A.M.; data curation, R.K.N. and M.A.M.; writing—original draft preparation, R.K.N.; writing—review and editing, K.Y., M.A.M., and R.K.N.; visualization, R.K.N.; project administration, P., K.Y., and M.A.M.; funding acquisition, P., K.Y., and M.A.M. All authors have read and agreed to the published version of the manuscript.

Funding

This research was funded by the Directorate of Research and Community Service (DPPM), Ministry of Higher Education, Science and Technology, Republic of Indonesia, through the BIMA Grant Program under the Community Partnership Empowerment Scheme (Skema Pemberdayaan Kemitraan Masyarakat) (Master Grant No.: 161/C3/DT.05.00/PM/2026; Universitas Mataram Derivative Grant No.: 3968/UN18.L1/PP/2026).

Informed Consent Statement

Informed consent was obtained from all participants, who voluntarily agreed that questionnaire and training-evaluation data could be used for research and publication. No invasive procedures or sensitive personal information were collected.

Data Availability

The original questionnaire forms (Bahasa Indonesia) and the Python scripts used for the statistical analyses reported in this study are deposited in the Zenodo repository and are publicly available at https://doi.org/10.5281/zenodo.21315043.

Acknowledgements

The authors thank the participating members of the Bina Mandiri Farmer Group and the farming community of Batu Mekar Village, the Batu Mekar Village Government for administrative support, and the Lingsar Agricultural Extension Office (BPP), West Lombok Department of Agriculture, for technical assistance. The authors also thank the colleagues, students, and field assistants from Universitas Mataram who assisted with the preparation, implementation, documentation, and evaluation of the training program.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used generative AI and AI-assisted technologies solely to improve English grammar, spelling, clarity, and style. All suggestions were reviewed and verified by the authors, who accept full responsibility for the manuscript's accuracy, integrity, and originality.

References

Anderson, J. R. & Feder, G. (2007). Agricultural extension. In *Handbook of Agricultural Economics* (Vol. 3, pp.

2343–2378). Elsevier. https://doi.org/10.1016/S1574-0072(06)03044-1.
Calvo, P., Nelson, L., & Kloepper, J. W. (2014). Agricultural uses of plant biostimulants. *Plant Soil*, *383*(1–2), 3–41. https://doi.org/10.1007/s11104-014-2131-8.
Christens, B. D. (2012). Targeting empowerment in community development: A community psychology approach to enhancing local power and well-being. *Community Dev. J.*, *47*(4), 538–554. https://doi.org/10.1093/cdj/bss031.
Davis, K., Nkonya, E., Kato, E., Mekonnen, D. A., Odendo, M., Miiro, R., & Nkuba, J. (2012). Impact of farmer field schools on agricultural productivity and poverty in East Africa. *World Dev.*, *40*(2), 402–413. https://doi.org/10.1016/j.worlddev.2011.05.019.
du Jardin, P. (2015). Plant biostimulants: Definition, concept, main categories and regulation. *Sci. Hortic.*, *196*, 3–14. https://doi.org/10.1016/j.scienta.2015.09.021.
Ensor, J. & Harvey, B. (2015). Social learning and climate change adaptation: Evidence for international development practice. *WIREs Clim. Change*, *6*(5), 509–522. https://doi.org/10.1002/wcc.348.
Firmansyah, A., Sumardjo, Fatchiya, A., & Sadono, D. (2024). Empowering environmentally friendly farmer communities: Social innovation to support sustainable agriculture. *IOP Conf. Ser.: Earth Environ. Sci.*, *1359*, 012050. https://doi.org/10.1088/1755-1315/1359/1/012050.
Goenaga, R. & Rivera, E. (2005). Growth and nutrient uptake of mangosteen grown under shade levels. *J. Agric. Univ. P.R.*, *89*(3–4), 149–158. https://doi.org/10.46429/jaupr.v89i3-4.1036.
Joshi, A., Kale, S., Chandel, S., & Pal, D. K. (2015). Likert scale: Explored and explained. *Br. J. Appl. Sci. Technol.*, *7*(4), 396–403. https://doi.org/10.9734/BJAST/2015/14975.
Karbo, R. T., Frewer, L. J., Areal, F., Jones, G., & Nurudeen, S. (2024). A systematic review of the efficacy of theories used to understand farmers' technology adoption behavior in lower-to-middle-income countries. *Dev. Stud. Res.*, *11*(1), 2294696. https://doi.org/10.1080/21665095.2023.2294696.
Klerkx, L., van Mierlo, B., & Leeuwis, C. (2012). Evolution of systems approaches to agricultural innovation: Concepts, analysis and interventions. In *Farming Systems Research Into the 21st Century: The New Dynamic* (pp. 457–483). Springer, Dordrecht. https://doi.org/10.1007/978-94-007-4503-2_20.
Knowler, D. & Bradshaw, B. (2007). Farmers' adoption of conservation agriculture: A review and synthesis of recent research. *Food Policy*, *32*(1), 25–48. https://doi.org/10.1016/j.foodpol.2006.01.003.
Kolb, D. A. (1984). *Experiential Learning: Experience as the Source of Learning and Development*. Prentice Hall.
Labonté, R. & Laverack, G. (2001). Capacity building in health promotion, part 1: For whom? And for what purpose? *Crit. Public Health*, *11*(2), 111–127. https://doi.org/10.1080/09581590110039838.
Laverack, G. (2006). Improving health outcomes through community empowerment: A review of the literature. *J. Health Popul. Nutr.*, *24*(1), 113–120.
Meijer, S. S., Catacutan, D., Ajayi, O. C., Sileshi, G. W., & Nieuwenhuis, M. (2015). The role of knowledge, attitudes and perceptions in the uptake of agricultural and agroforestry innovations among smallholder farmers in sub-Saharan Africa. *Int. J. Agric. Sustain.*, *13*(1), 40–54. https://doi.org/10.1080/14735903.2014.912493.
Nazre, M. (2014). New evidence on the origin of mangosteen (*Garcinia mangostana* L.) based on morphology and ITS sequence. *Genet. Resour. Crop Evol.*, *61*(6), 1147–1158. https://doi.org/10.1007/s10722-014-0097-2.
Nor Diana, M. I., Zulkepli, N. A., Ern, L. K., & Zainol, M. R. (2024). Factors affecting behavioral intentions of farmers in Southeast Asia to technology adoption: A systematic review analysis. *J. Environ. Manage.*, *367*, 122045. https://doi.org/10.1016/j.jenvman.2024.122045.
Nugroho, B. D. A., Nurrohman, R. K., & Sudira, P. (2020). Relationships among global climate indices and rainfall pattern to detect impact of climate change in Yogyakarta Special Region, Indonesia. *IOP Conf. Ser.: Earth Environ. Sci.*, *515*, 012038. https://doi.org/10.1088/1755-1315/515/1/012038.
Osman, M. B. & Milan, A. R. (2006). *Mangosteen: Garcinia mangostana L.* University of Southampton. https://eprints.soton.ac.uk/53412/
Pithakpol, W., Sawatsuk, B., Tana, S., Khamsamut, Y., & Sirisathitkul, C. (2025). Sustainable agriculture network of organic rice production: The model of community enterprise in Phayao Province, Thailand. *Int. J. Agric. Sci. Res. Technol. Ext. Educ. Syst.*, *15*(3), 151–159. https://doi.org/10.71505/ijasrt.2025.1209792.
Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *J. Appl. Psychol.*, *88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879.
Pretty, J. N. (1995). Participatory learning for sustainable agriculture. *World Dev.*, *23*(8), 1247–1263. https://doi.org/10.1016/0305-750X(95)00046-F.
Reed, M. S., Evely, A. C., Cundill, G., Fazey, I., Glass, J., Laing, A., Newig, J., Parrish, B., Prell, C., Raymond, C. et al. (2010). What is social learning? *Ecol. Soc.*, *15*(4), r1. https://doi.org/10.5751/ES-03564-1504r01.
Rogers, E. M. (2003). *Diffusion of Innovations (5th ed.)*. Free Press.

Rouphael, Y. & Colla, G. (2020). Toward a sustainable agriculture through plant biostimulants: From experimental data to practical applications. *Agronomy*, *10*(10), 1461. https://doi.org/10.3390/agronomy10101461.
Ruzzante, S., Labarta, R., & Bilton, A. (2021). Adoption of agricultural technology in the developing world: A meta-analysis of the empirical literature. *World Dev.*, *146*, 105599. https://doi.org/10.1016/j.worlddev.2021.105599.
Scoones, I. (1998). *Sustainable rural livelihoods: A framework for analysis* (IDS Working Paper No. 72). Brighton, UK. https://hdl.handle.net/20.500.12413/3390.
Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin.
Shireen, F., Nawaz, M. A., Chen, C., Zhang, Q., Zheng, Z., Sohail, H., Sun, J., Cao, H., Huang, Y., & Bie, Z. (2018). Boron: Functions and approaches to enhance its availability in plants for sustainable agriculture. *Int. J. Mol. Sci.*, *19*(7), 1856. https://doi.org/10.3390/ijms19071856.
Sukayat, Y., Setiawan, I., Suharfaputra, U., & Kurnia, G. (2023). Determining factors for farmers to engage in sustainable agricultural practices: A case from Indonesia. *Sustainability*, *15*(13), 10548. https://doi.org/10.3390/su151310548.
Wang, M., Zheng, Q., Shen, Q., & Guo, S. (2013). The critical role of potassium in plant stress response. *Int. J. Mol. Sci.*, *14*(4), 7370–7390. https://doi.org/10.3390/ijms14047370.
Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biom. Bull.*, *1*(6), 80–83. https://doi.org/10.2307/3001968.
Yao, T. L., Nazre, M., McKey, D., Jalonen, R., & Duminil, J. (2023). The origin of cultivated mangosteen (*Garcinia mangostana* L. var. *mangostana*): Critical assessments and an evolutionary-ecological perspective. *Ecol. Evol.*, *13*(3), e9792. https://doi.org/10.1002/ece3.9792.
Zörb, C., Senbayram, M., & Peiter, E. (2014). Potassium in agriculture—Status and perspectives. *J. Plant Physiol.*, *171*(9), 656–669. https://doi.org/10.1016/j.jplph.2013.08.008.

Appendix

A. Supplementary Tables

Table A1. Exploratory item-level results for generative-fertilization perceived knowledge

Code	Item	Cohort	*n*	Pre	Post	Mean Change	W^{+}	Exact *p*	r_{rb}
GF1	I understand the difference between vegetative and generative fertilization.	Prospective	7	3.14 ± 0.38	5.00 ± 0.00	+1.86	28	0.016	1.00
GF1	I understand the difference between vegetative and generative fertilization.	Retrospective	14	3.14 ± 0.36	4.71 ± 0.47	+1.57	105	<0.001	1.00
GF2	I know the roles of phosphorus (P) and potassium (K) in flowering and fruit set.	Prospective	7	2.86 ± 0.69	5.00 ± 0.00	+2.14	28	0.016	1.00
GF2	I know the roles of phosphorus (P) and potassium (K) in flowering and fruit set.	Retrospective	14	2.86 ± 0.66	4.71 ± 0.47	+1.86	105	<0.001	1.00
GF3	I understand the proper timing for applying generative fertilizer.	Prospective	7	2.71 ± 0.49	5.00 ± 0.00	+2.29	28	0.016	1.00
GF3	I understand the proper timing for applying generative fertilizer.	Retrospective	14	2.71 ± 0.47	4.71 ± 0.47	+2.00	105	<0.001	1.00
GF4	I understand the expected benefits of generative fertilization on fruit production.	Prospective	7	3.00 ± 0.58	5.00 ± 0.00	+2.00	28	0.016	1.00
GF4	I understand the expected benefits of generative fertilization on fruit production.	Retrospective	14	3.00 ± 0.55	5.00 ± 0.00	+2.00	105	<0.001	1.00

Note: n = sample size. W^{+} denotes the sum of positive Wilcoxon signed ranks; r_{rb} denotes the matched-pairs rank-biserial correlation. Exact p-values are two-sided and unadjusted. No multiplicity adjustment was applied; item-level results are exploratory.

Table A2. Exploratory item-level results for fruiting-booster perceived knowledge

Code	Item	Cohort	*n*	Pre	Post	Mean Change	W^+	Exact *p*	r_{rb}
FB1	I understand the role of flowering enhancers in stimulating flowering.	Prospective	7	2.86 ± 0.38	5.00 ± 0.00	+2.14	28	0.016	1.00
FB1	I understand the role of flowering enhancers in stimulating flowering.	Retrospective	14	3.00 ± 0.55	4.71 ± 0.47	+1.71	105	<0.001	1.00
FB2	I know the correct application methods for flowering enhancers.	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	1.00
FB2	I know the correct application methods for flowering enhancers.	Retrospective	14	3.14 ± 0.36	4.57 ± 0.51	+1.43	105	<0.001	1.00
FB3	I understand the risks associated with excessive use of flowering enhancers.	Prospective	7	2.71 ± 0.49	5.00 ± 0.00	+2.29	28	0.016	1.00
FB3	I understand the risks associated with excessive use of flowering enhancers.	Retrospective	14	3.00 ± 0.55	4.71 ± 0.47	+1.71	105	<0.001	1.00
FB4	I know the environmental conditions that affect the success of flowering.	Prospective	7	2.57 ± 0.53	5.00 ± 0.00	+2.43	28	0.016	1.00
FB4	I know the environmental conditions that affect the success of flowering.	Retrospective	14	2.86 ± 0.66	4.71 ± 0.47	+1.86	105	<0.001	1.00
FB5	I understand how flowering enhancers can affect the yield and quality of mangosteen.	Prospective	7	2.86 ± 0.38	5.00 ± 0.00	+2.14	28	0.016	1.00
FB5	I understand how flowering enhancers can affect the yield and quality of mangosteen.	Retrospective	14	3.00 ± 0.55	4.71 ± 0.47	+1.71	105	<0.001	1.00

Note: *n* = sample size. W^+ denotes the sum of positive Wilcoxon signed ranks; r_{rb} denotes the matched-pairs rank-biserial correlation. Exact *p*-values are two-sided and unadjusted. No multiplicity adjustment was applied; item-level results are exploratory.

Table A3. Exploratory item-level results for readiness-related responses

Code	Item	Cohort	*n*	Pre	Post	Mean Change	W^+	Exact *p*	r_{rb}
AR1	I am confident that I can apply the techniques introduced on my own.	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	1.00
AR1	I am confident that I can apply the techniques introduced on my own.	Retrospective	14	3.14 ± 0.36	4.71 ± 0.47	+1.57	105	<0.001	1.00
AR2	I am willing to apply this technology in my orchard.	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	1.00
AR2	I am willing to apply this technology in my orchard.	Retrospective	14	3.14 ± 0.36	4.86 ± 0.36	+1.71	105	<0.001	1.00
AR3	I am confident that this technology is economically viable.	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	1.00
AR3	I am confident that this technology is economically viable.	Retrospective	14	3.14 ± 0.36	4.57 ± 0.51	+1.43	105	<0.001	1.00
AR4	I plan to recommend this technology to other farmers.	Prospective	7	3.00 ± 0.00	5.00 ± 0.00	+2.00	28	0.016	1.00
AR4	I plan to recommend this technology to other farmers.	Retrospective	14	3.14 ± 0.36	4.57 ± 0.51	+1.43	105	<0.001	1.00

Note: *n* = sample size.; W^+ denotes the sum of positive Wilcoxon signed ranks; r_{rb} denotes the matched-pairs rank-biserial correlation. Exact *p*-values are two-sided and unadjusted. No multiplicity adjustment was applied; item-level results are exploratory.

Table A4. Sign-test sensitivity analyses for the six principal comparisons

Outcome domain	Cohort	*n*	Positive changes	Negative changes	Ties	Exact *p*
Generative Fertilization	Prospective	7	7	0	0	0.016
Generative Fertilization	Retrospective	14	14	0	0	<0.001
Fruiting Booster	Prospective	7	7	0	0	0.016
Fruiting Booster	Retrospective	14	14	0	0	<0.001
Readiness-related responses	Prospective	7	7	0	0	0.016
Readiness-related responses	Retrospective	14	14	0	0	<0.001

Note: Ties denote zero paired differences. Exact two-sided sign-test *p*-values are unadjusted and are reported as sensitivity analyses only. *n* = sample size.

Appendix B. Retained blank de-identified analytical questionnaire (Bahasa Indonesia and English)

1. Bahasa Indonesia

Bagian A. Karakteristik Peserta

Nama:
Jenis kelamin: Laki-laki / Perempuan
Usia (tahun):
Tingkat Pendidikan:
Pengalaman bertani (tahun):
Luas lahan manggis (ha):
Jumlah pohon manggis yang produktif:
Produksi tahunan rata-rata (kg/tahun):
Apakah Anda pernah mengikuti pelatihan serupa sebelumnya? (Ya/Tidak)

Bagian B. Penilaian Pengetahuan yang Dirasakan (Sebelum dan Setelah Pelatihan) Skala:

1 = Sangat Tidak Setuju
2 = Tidak Setuju
3 = Netral
4 = Setuju
5 = Sangat Setuju

B1. Pengetahuan tentang Pupuk Generatif

Kode	Indikator	Skor Sebelum	Skor Sesudah
GF1	Saya memahami perbedaan antara pemupukan vegetatif dan generatif.		
GF2	Saya mengetahui peran fosfor (P) dan kalium (K) dalam pembungaan dan pembentukan buah.		
GF3	Saya memahami waktu yang tepat untuk aplikasi pupuk generatif.		
GF4	Saya memahami manfaat yang diharapkan dari pemupukan generatif terhadap produksi buah.		

B2. Pengetahuan tentang Teknologi Peningkat Pembungaan

Kode	Indikator	Skor Sebelum	Skor Sesudah
FB1	Saya memahami fungsi peningkat pembungaan dalam merangsang pembungaan.		
FB2	Saya mengetahui metode aplikasi yang benar untuk penguat pembuahan.		
FB3	Saya memahami risiko yang terkait dengan aplikasi penguat pembuahan yang berlebihan.		
FB4	Saya mengetahui kondisi lingkungan yang memengaruhi keberhasilan pembungaan.		
FB5	Saya memahami bagaimana penguat pembuahan dapat memengaruhi hasil dan kualitas manggis.		

B3. Kesiapan Penerapan

Kode	Indikator	Skor Sebelum	Skor Sesudah
AR1	Saya yakin dapat menerapkan teknik-teknik yang diperkenalkan secara mandiri.		
AR2	Saya bersedia menerapkan teknologi ini di kebun saya.		
AR3	Saya yakin teknologi ini layak secara ekonomi.		
AR4	Saya berencana merekomendasikan teknologi ini kepada petani lain.		

Bagian C. Evaluasi Pelatihan (Setelah Pelatihan)

Kode	Indikator	Skor
TE1	Materi pelatihan mudah dipahami.	
TE2	Para instruktur menjelaskan materi dengan jelas.	
TE3	Demonstrasi praktis meningkatkan pemahaman saya.	
TE4	Pelatihan ini relevan dengan kegiatan bertani saya.	
TE5	Secara keseluruhan, saya puas dengan program pelatihan ini.	

2. English Version

Section A. Participant Characteristics

Name:
Gender: Male / Female
Age (years) :
Education Level:
Farming Experience (years):
Mangosteen Cultivation Area (ha):
Number of Productive Mangosteen Trees:
Average Annual Production (kg/year) :
Have you previously participated in similar training? (Yes/No)

Section B. Perceived Knowledge Assessment (Before and After Training) Scale:

1 = Strongly Disagree
2 = Disagree
3 = Neutral
4 = Agree
5 = Strongly Agree

B1. Knowledge of Generative Fertilization

Code	Indicator	Pre-Training	Post-Training
GF1	I understand the difference between vegetative and generative fertilization.		
GF2	I understand the role of phosphorus (P) and potassium (K) in flowering and fruit formation.		
GF3	I understand the appropriate timing for applying generative fertilizer.		
GF4	I understand the expected benefits of generative fertilization for fruit production.		

B2. Knowledge of Fruiting Booster Technology

Code	Indicator	Pre-Training	Post-Training
FB1	I understand the function of fruiting boosters in stimulating flowering.		
FB2	I know the correct application method for the fruiting booster.		
FB3	I understand the risks associated with excessive application of the fruiting booster.		
FB4	I know the environmental conditions that influence successful flowering.		
FB5	I understand how the fruiting booster can affect mangosteen yield and fruit quality.		

B3. Adoption Readiness

Code	Indicator	Pre-Training	Post-Training
AR1	I am confident that I can independently apply the techniques introduced during the training.		
AR2	I am willing to implement this technology in my orchard.		
AR3	I believe this technology is economically feasible.		
AR4	I intend to recommend this technology to other farmers.		

Section C. Training Evaluation (After Training)

Code	Indicator	Scale
TE1	The training materials were easy to understand.	
TE2	The instructors explained the materials clearly.	
TE3	The practical demonstrations improved my understanding.	
TE4	The training was relevant to my farming activities.	
TE5	Overall, I am satisfied with the training program.	